Nutrition 77 (2020) 110833

Contents lists available at ScienceDirect

Nutrition

journal homepage: www.nutritionjrnl.com

Applied nutritional investigation

# Utility of bioimpedance methods for the assessment of fat-free mass in adult outpatients with inflammatory bowel disease

Jessica Wood M.Clin.Res. [a,*], Leigh Ward Ph.D., F.N.S.A. [b], Miles Sparrow M.B.B.S., F.R.A.C.P. [c,d], Susannah King Ph.D., M.P.H., M.Nut.Diet [a,e]

[a] Nutrition Department, Alfred Hospital, Melbourne VIC, Australia
[b] School of Chemistry and Molecular Biosciences, The University of Queensland, St Lucia, QLD, Australia
[c] Department of Gastroenterology, Alfred Hospital, Melbourne VIC, Australia
[d] Department of Medicine, Nursing and Health Sciences, Monash University, The Alfred Centre, Melbourne VIC, Australia
[e] Department of Dietetics, Nutrition and Sport, La Trobe University, Bundoora VIC, Australia

ARTICLE INFO

*Article History:*
Received 28 January 2020
Received in revised form 15 March 2020
Accepted 17 March 2020

*Keywords:*
Inflammatory bowel disease
Fat-free mass
Bioimpedance spectroscopy
Multifrequency bioimpedance analysis
Dual energy x-ray absorptiometry
Body composition

ABSTRACT

*Objectives:* Clinically accessible body composition assessment is required to identify fat-free mass (FFM) depletion, which is common in inflammatory bowel disease (IBD) and poorly correlated with body mass index (BMI). The aim of this study was to compare FFM assessed using bioimpedance spectroscopy (BIS) and multifrequency bioelectrical impedance analysis (MFBIA) with dual energy x-ray absorptiometry (DXA) in adult outpatients with IBD.
*Methods:* FFM was measured using DXA, BIS, and MFBIA in 40 outpatients with IBD. Paired *t* tests, Lin and Pearson's correlations, and limits of agreement (LOA) analysis were used to compare $FFM_{DXA}$ with $FFM_{BIS}$ and $FFM_{MFBIA}$.
*Results:* Participants (26 men, 24 with Crohn's disease) had a median (IQR) age 39 y (32–50 y) and median (IQR) BMI 24.2 kg/m$^2$ (21.9–26.4 kg/m$^2$). Mean $FFM_{DXA}$ was 51.6 ± 12 kg. $FFM_{BIS}$ was highly correlated with $FFM_{DXA}$ ($r_p$ = 0.97; $P < 0.001$); however, BIS significantly overestimated FFM compared with DXA by a mean 3.4 ± 2.6 kg ($P < 0.001$) and showed wide 95% LOA (−1.7 to 8.4 kg). $FFM_{BIS}$ estimations improved when FFM was adjusted for by BMI using Moissl's predictive algorithm, reducing mean bias to 0.1 ± 2 kg ($P$ = 0.858; 95% LOA −3.9 to 4 kg). The bias between $FFM_{MFBIA}$ and $FFM_{DXA}$ was 1.3 ± 2 kg ($P$ = 0.002) with 95% LOA −2.6 to 5 kg (n = 28).
*Conclusion:* Unadjusted BIS overestimated FFM in IBD outpatients compared with DXA with poor agreement at an individual level. Adjusting for BMI ameliorated the overestimation. It is suggested for the estimation of FFM in outpatients with IBD that MFBIA or the Moissl algorithm with BIS be used if DXA is unavailable.
© 2020 Elsevier Inc. All rights reserved.

## Introduction

Inflammatory bowel disease (IBD) is a chronic, immune-mediated, inflammatory, relapsing and remitting disease that results in inflammation and ulceration along any site of the gastrointestinal tract [1]. IBD has three subtypes: Crohn disease (CD), ulcerative colitis (UC), and inflammatory bowel disease unspecified (IBD-U).

Baseline differences in body composition in patients with IBD, like in all populations, can be due to sex, ethnicity, or lifestyle. Complications of IBD include malnutrition and unintentional weight loss, which alters body composition as a result of loss of fat-free mass (FFM) and fat mass. Anorexia, increased energy expenditure, malabsorption, and food exclusions contribute to malnutrition in IBD [2,3]. Recent studies have shown between 22% and 60% of patients with IBD have FFM depletion [4–7]. This is significant because FFM depletion has been associated with negative outcomes including major postoperative complications, predicting a small bowel resection, primary non-response to antitumor necrosis factor (TNF) agents, and osteopenia [4,7–9].

Traditional nutritional measurements, like body mass index (BMI), correlate poorly with indices of FFM in patients with CD [4,9] resulting in a risk for underrecognition and underestimation of the extent

An Alfred Health small project grant (no. T1170) provided partial support for undertaking this study. No other financial support was received. JW was primarily responsible for the study design and implemented the study including data collection, management, and analysis of data. SK provided significant input into the study design, implementation, data, and statistical analysis. MS served as the study physician, and provided significant advice/consultation in study planning and implementation. LW completed data analysis for the Moissl method and equivalence testing and provided advice in the statistical analysis. All authors critically revised the manuscript, read and approved the final manuscript, and agree to be fully accountable for ensuring the integrity and accuracy of the work. The authors have no conflicts of interests to declare.
*Corresponding author: Tel.: ± 61 613 9903 0233.
*E-mail address:* Jessica.wood@alfred.org.au (J. Wood).

https://doi.org/10.1016/j.nut.2020.110833
0899-9007/© 2020 Elsevier Inc. All rights reserved.

of nutrition depletion when relying only on weight-based indicators of nutritional status. The European Society of Enteral and Parental Nutrition (ESPEN) introduced body composition into its malnutrition definition, recognizing that nutritional depletion may be missed with weight-based indicators alone and that FFM depletion can occur in normal, overweight, and obese populations [10]. Reduced muscle mass has been included in the Global Consensus for Diagnosing Malnutrition in Adult Patients [11]. Therefore reliable, affordable, and accessible body composition tools need to be available for clinical populations to assist in identifying FFM depletion and evaluating response to nutritional interventions.

Dual energy x-ray absorptiometry (DXA) is a reliable and precise technique for assessing body composition, and has been widely used as a reference method. However, DXA involves exposure, albeit small, to ionizing radiation, may not be accessible at the clinic point-of-care, and the cost is not rebatable in many health systems. Therefore, DXA is limited in utility in the clinical setting for outpatient body composition assessment.

Despite multiple reports of body composition measurements in IBD, there are few studies, to our knowledge, that compare different methods to ascertain which are sufficiently reliable and accurate for clinical use, and none comparing bedside techniques to a reference method, in either inpatient or outpatient IBD settings. Bioimpedance (BIA) methodologies that are non-invasive, affordable, and accessible in the clinical setting have not been well validated in outpatient IBD populations. Despite this, BIA is appearing in IBD clinical trials as the method for body composition [12–14]. It is important to assess the utility of BIA for the outpatient IBD population to ensure that researchers and clinicians are using methods that produce sufficiently accurate body composition results that help them understand the patterns of altered body composition, its risk factors and consequences, and to monitor the effects of interventions.

The aim of this study was to compare bedside body composition assessment tools, bioimpedance spectroscopy (BIS) and multifrequency bioimpedance analysis (MFBIA), with the reference technique (DXA) to determine if these tools can provide an accurate assessment of FFM for adult IBD outpatients.

## Materials and methods

### *Participants*

Forty adult participants were recruited sequentially from IBD outpatient clinics at Alfred Health, Melbourne, between December 2017 and June 2018. Inclusion criteria included ≥20 y of age and a confirmed diagnosis of IBD on histologic, endoscopic, and clinical criteria [1]. Exclusion criteria included too heavy (>120 kg) or too tall (>190 cm) for the DXA scanner; permanent cardiac-implanted device precluding BIA measurements, pregnancy or breastfeeding, CD confined to perianal distribution, or inability to give informed consent or complete the study requirements (intellectual disability, insufficient English-language proficiency). The study was approved by Alfred Health Ethics Committee. Informed formal written consent was obtained from participants before participation. All procedures performed in this study that involved human participants were in accordance with the ethical standards of the Alfred Hospital and with the 1964 Helsinki Declaration and its later amendments.

### *Study protocol*

All data and measurements were collected at one study visit except for the dietary assessment, which was completed at home. Participants refrained from consuming food, alcohol, and caffeine in the 8 h before assessment, were advised to come hydrated, avoid strenuous exercise that day, and void within the 30 min before assessment.

Demographic and clinical information included age, sex, current medical history, past medical and surgical history, use of steroids within the previous 3 mo, current medications and C-reactive protein (CRP; within the previous month) were obtained from electronic medical records. Disease activity scores were calculated using the Harvey Bradshaw Index for CD and Partial Mayo score for UC and IBD-U [15,16]. Disease phenotype was classified by Montreal criteria [17]. Weight and height were measured with partiicpants dressed in light clothing, shoes removed and after voiding, using electronic scales and a wall-mounted stadiometer to the nearest 0.1 kg and 0.1 cm, respectively. BMI was calculated by dividing weight in kilograms (kg) by height (meters squared) and classified per the World Health Organization [18].

### *Body composition assessment: BIS, MFBIA, and DXA*

#### *BIS*

The BIS device (ImpSFB7, ImpediMed Ltd., Brisbane, Australia) was calibrated before each use via its inbuilt calibration function. Participants lay supine on a hospital bed for 10 min before BIS measurement. Sites were cleaned with an alcohol swab before affixing Ag-AgCl gel electrocardiogram-style electrodes, which were placed on the right hand, wrist, ankle, and foot using conventional anatomic locations as recommended by the manufacturer.

The ImpSFB7 BIS device passes a low <200 μA, alternating electrical current through the participant scanning over 256 frequencies between 3 and 1000 kHz to determine measures of resistance and reactance used to calculate total body water (TBW), extracellular water (ECW) and intracellular water (ICW) from using Cole modeling and Hanai mixture theory (Bioimp v 4.18.0.0, ImpediMed Ltd.). For Hanai modeling, the device's default resistivity constants were used (ρ female 235.5, ρ female 894.2, ρ male 273.9 ρ male 937.2) with body density 1.05 kg/L and body proportion (kb) 4.3 [19].

FFM from the BIS ($FFM_{BIS}$) was calculated by the device by dividing TBW by a hydration fraction of 0.732 mL/g [19]. Three wrist-to-ankle measurements were taken at 5-min intervals (without electrode replacement) with the average of the three measurements of each of TBW, ECW, intracellular water, FFM, and fat mass (FM) recorded.

Data were also analyzed using resistivity coefficients based on an Australian population [19]. The Moissl modification to the Hanai model was also applied to the data, which attempts to account for body shape by adjusting for BMI [20].

#### *MFBIA*

Participants were dressed in light clothing with shoes and socks removed and stood on the seca 515 MFBIA device (Hamburg, Germany). Measurements were obtained between the hands and feet using an 8-electrode arrangement with the participant standing on sets of electrodes while gripping the handrail electrodes. An electrical current was passed between electrodes and the device measured resistance and reactance at 20 discrete frequencies in the range of 1 to 1000 kHz. FFM from the BIS ($FFM_{MFBIA}$) was calculated by the device.

#### *DXA*

Whole-body DXA scans were performed using a Lunar Prodigy (GE Lunar Corp, Madison, WI, USA) according to standardized procedures recommended by the manufacturer with calibration performed daily. Participants were scanned while dressed in a gown, and scans were analyzed using the manufacturer's software (version 13:60:033). $FFM_{DXA}$ (kg) was calculated as the sum of the estimates of lean tissue mass and bone mineral content.

### *Sample size and power calculation*

Sample size calculation was based on a previous study in which FFM had a negligible mean bias (0.24 kg) between DXA and BIS analysis in a healthy population [21]. There is no internationally accepted minimum clinically important difference (MCID) between body composition techniques and for the BIS to be a useful clinical tool it would need to correctly classify FFM depletion using the ESPEN definition, therefore a clinically significant a priori threshold was set of 1.5 kg (with SD 2.5 kg) as the MCID FFM at a group level between DXA and each BIA method [10]. A mean bias of 1.5 kg FFM equates to ~0.5 units of FFMI. The power calculation used a predefined available sample size for the study (n = 40 for BIS and n = 28 for MFBIA) and a two-sided α of 0.05. The power calculation demonstrated a sample size of 40 had a power of 95% to detect an MCID of 1.5 kg and the sample of 28 for MFBIA had a power of 89%.

### *Statistical analysis*

Data analysis was undertaken using SPSS version 26 (IBM, Chicago, IL, USA). Data are presented as mean ± SD and median (interquartile range [IQR]) for variables according to the distribution. Pearson's and Lin's concordance correlations were used to determine the linear correlation between $FFM_{DXA}$ and each of the bioimpedance techniques. Paired *t* tests were used to determine if there were statistically significant differences between $FFM_{DXA}$ and FFM using bioimpedance methods. Two SD LOA analysis, using Bland–Altman analysis, and concordance correlation were undertaken to quantify the agreement between the bioimpedance methods and DXA [22]. To investigate accuracy at an individual level, the proportion of participants with a FFM result by each bioimpedance method that was within 5% of their $FFM_{DXA}$ was used [23]. Equivalence testing was completed using a two one-sided *t* test procedure. Statistical significance was set at $P < 0.05$.

**Table 1**
Participant demographic and clinical information

| | CD | UC | IBD-U | All |
|---|---|---|---|---|
| Patients, n (%) | 24 (60) | 12 (30) | 4 (10) | 40 |
| Men, n (%) | 14 (58) | 7 (58) | 3 (75) | 24 (60) |
| Age (y), median (IQR) | 38 (30–47) | 36 (32–56) | 43 (33–51) | 39 (32–50) |
| Ethnicity (white), n (%) | 22 (92) | 11 (85) | 4 (100) | 37 (93) |
| IBD information | | | | |
| Disease duration (y), median (IQR) | 11 (3–18) | 2 (1–12) | 7 (6–10) | 7 (1–15) |
| CRP (mg/L), median (IQR)* | 3 (3–10) | 3(3–12) | 3 (3–6.5) | 3(3–13) |
| Clinical disease score (HBI, PMS), median (IQR) | 0 (0–5) | 1 (0–2) | 1 (0–4) | |
| IBD-related surgery (yes), n (%) | 6(26) | 0(0) | 0(0) | 6(15) |
| Patients with ileostomy, n (%) | 2(%) | 0(0) | 0(0) | 2(5) |
| Steroid use in previous 3 mo (yes), n (%) | 2(9) | 4(33) | 1(25) | 7(18) |

CD, Crohn disease; CRP, C- reactive protein; HBI, Harvey–Bradshaw index; IBD, inflammatory bowel disease; IBD-U, inflammatory bowel disease unspecified; IQR, interquartile range; PMS, partial Mayo score; UC, ulcerative colitis
*NB missing data for age at diagnosis and CRP in one and four participants, respectively

## Results

Twenty-six men and 14 women participated (median age 39 y [IQR 32–50 y]). Demographic and clinical characteristics are presented in Table 1 and Supplementary Table 1. Of the participants, 40% (n = 16) were receiving biological therapy and 68% (n = 27) had previous dietitian intervention as part of attending the outpatient clinic.

### *Body composition determined by reference method*

Body composition results for the reference method DXA are detailed in Table 2. Mean BMI did not differ between men and women. Of the participants, 35% (n = 14) were either overweight (22%, n = 9) or obese (13%, n = 5).

### *Hydration status*

Participants had hydration parameters within normal limits ($ECW_{BIS}$ %/$TBW_{BIS}$ % median, IQR): men 0.42 L (0.41–0.44), women

**Table 2**
Body composition values from DXA

| | Men (n = 24) | Women (n = 16) | All (N = 40) |
|---|---|---|---|
| Height (m)† | 1.79 ± 0.08 | 1.63 ± 0.08 | 1.72 ± 0.11 |
| Weight (kg)† | 79.1 ± 12.1 | 66 ± 16 | 73.8 ± 15 |
| BMI (kg/m$^2$) (median, IQR) | 24.3 (22.8–26.1) | 24 (21.4–27.7) | 24.8 (21.9–26.4) |
| BMI >30 kg/m$^2$, % (n) | 8 (2) | 19 (3) | 13 (5) |
| $FFM_{DXA}$(kg)* | 59.6 ± 7.7 | 39.6 ± 5.4 | 51.6 ± 12 |
| $FFMI_{DXA}$ (kg/m$^2$)* | 18.8 ± 1.8 | 14.8 ± 0.9 | 17.2 ± 2.5 |
| $FM_{DXA}$(kg)* | 19.3 ± 7 | 26.4 ± 11.7 | 22.2 ± 9.7 |
| $FM\%_{DXA}$* | 24.1 ± 6.4 | 38.4 ± 8.6 | 29.8 ± 10.1 |
| $FMI_{DXA}$(kg/m$^2$)* | 6.1 ± 2.2 | 9.8 ± 3.9 | 7.6 ± 3.4 |

BMI, body mass index; DXA, dual energy x-ray absorptiometry; $FFM_{DXA}$, fat-free mass from DXA; FFMI, fat-free mass index; $FFMI_{DXA}$, fat-free mass index from DXA; $FM\%_{DXA}$, fat mass percentage from DXA;
Data shown as mean ± SD except where noted
*Men vs women $P < 0.05$.
†Median and interquartile range.

**Table 3**
Comparison of bedside body composition methods to DXA in determining FFM

| | FFM (kg) Mean ± SD | Pearson's correlation (DXA vs method) r value | Lin's concordance correlation (DXA vs method) r value | Bias b/w DXA and method (kg) mean ± SD | Correlation between bias and mean (DXA and method) r value | 2-SD LOA (kg) | 2-SD LOA (%) | Equivalence (%) |
|---|---|---|---|---|---|---|---|---|
| $FFM_{DXA}$ | 51.6 ± 12 | NA | NA | NA | NA | NA | NA | NA |
| $FFM_{BIS}$ | 55 ± 12.5 | 0.98* | 0.91* | 3.4 ± 2.5 | 0.20 | −1.7 to 8.4 | 18.9 | 7.7 |
| $FFM_{BISAusRes}$ | 54.5 ± 12.9 | 0.98* | 0.92* | 2.9 ± 2.5* | 0.35* | −2.1 to 7.8 | 18.7 | 7.2 |
| $FFM_{BISMoisI}$ | 51.7 ± 11.8 | 0.99* | 0.99* | 0.1 ± 2 | 0.19 | −4 to 4 | 15.4 | 4.5 |
| $FFM_{MFBIA}$† | 52.5 ± 12 | 0.99* | 0.98* | 1.3 ± 2 | −0.19 | −2.6 to 5.1 | 13.5 | 6.9 |

BIS, bioimpedance spectroscopy; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass; LOA, limits of agreement;
MFBIA, multifrequency bioelectrical impedance analysis; BISAusRes, bioimpedance spectroscopy Australian reference population resistivity coefficients; NA, not applicable; b/w, between.
*Mean values reached statistical significance ($P < 0.05$).
†n = 28.

0.42 L (0.41−0.42). The comparisons of the BIA techniques with DXA are shown in Table 3.

*Performance of the BIS compared with reference method in determining FFM*

$FFM_{DXA}$ had a strong linear correlation and concordance correlation with FFM by all bioimpedance techniques (Table 3). When unadjusted, BIS overestimated FFM by an average of 3.4 kg compared with DXA exceeding the a priori threshold of a clinically important 1.5 kg mean bias ($P < 0.001$). On an individual level, 40% of participants (n = 16) had a $FFM_{BIS}$ within 5% of their $FFM_{DEXA}$. Twelve participants (30%) had an $FFM_{BIS}$ within 1.5 kg of $FFM_{DXA}$. There was no statistical difference between these two subgroups regarding age, disease type, sex, hydration ($ECW_{BIS}/TBW_{BIS}$), CRP, or disease score (data not shown).

The mean bias dropped below the MCID when FFM was adjusted for BMI using the approach of Moissl et al. ($FFM_{BISMoissl}$); however, despite a small improvement when using the Australian reference population resistivity coefficients ($FFM_{BISAusRes}$), it continued to exceed the MCID [19,20]. $FFM_{BISResAus}$ was the only method for which the bias increased as FFM increased ($r = 0.35$, $P = 0.03$; Table 3). As the BMI increased so did the bias between $FFM_{BIS}$ and $FFM_{DXA}$ ($r = 0.71$, $P < 0.0001$; Fig. 1).

*Performance of the MFBIA compared with reference method in determining FFM*

The mean bias between $FFM_{MFBIA}$ and $FFM_{DXA}$ was below the MCID at 1.3 kg. On an individual level, 64% of participants (n = 25) had an $FFM_{MFBIA}$ within 5% of their $FFM_{DEXA}$. When Bland−Altman's LOA analyses were used to compare the bias between FFM derived from the bioimpedance methods and $FFM_{DXA}$, all bioimpedance techniques had wide LOA varying from ± 3.6 kg for MFBIA to ± 5 kg for unadjusted BIS. In examining the similarities between measurements using equivalence testing between $FFM_{DXA}$ and $FFM_{BISMoissl}$, the measurements had an equivalence at 4.5%, better than the values for $FFM_{BIS,}$ $FFM_{BISAusRes}$, and $FFM_{BIA}$ (7.7%, 7.2%, and 6.9%, respectively). This means that >95% of the data from $FFM_{MoissI}$ for clinical and practical purposes are equivalent to DXA FFM measurements at an $\alpha = 0.05$ with 90% confidence intervals.

*Comparing ESPEN malnutrition criteria by methodology*

Four patients (10%) met the ESPEN malnutrition criteria element for depleted FFM using DXA. BIS correctly classified two of these. BIS did not give any false-positive results. In the 28 participants who had MFBIA, 2 were identified with FFM depletion using ESPEN criteria by DXA. MFBIA did not identify either of these with FFM depletion.

## Discussion

Despite widespread use of BIA in the literature, to our knowledge, this was the first study to compare bedside techniques to a reference standard for body composition assessment in IBD. The present study found that in adult outpatients with IBD, unadjusted BIS overestimated FFM compared with DXA by a mean of 3.4 kg, and this bias increased as BMI increased. The Moissl algorithm [20], which adjusts for an individual's BMI, reduced this overestimation of FFM, with a non-significant mean bias between FFM measured by DXA and BIS of 0.1 kg [20]. Clinicians who do not have access to BIS could use MFBIA (seca mBCA 515) to assess FFM in IBD outpatients (mean bias of 1.3 kg ± 2 kg).

This overestimation of FFM by unadjusted BIS and wide LOA have been reported in other populations. BIS overestimated mean FFM by 3 ± 2.5 kg (95% LOA −2.5 kg to 8.7 kg) and by 7.2 ± 3.8 kg in postpartum women and in athletes, respectively, when compared with DXA [24,25]. Conversely, BIS underestimated mean FFM compared with DXA by 7.6 ± 4.7 kg and 9.6 kg (SD not stated) in patients with advanced cancer and those with chronic obstructive pulmonafy disease, respectively [26,27].

The overestimation of FFM using unadjusted BIS is of concern because patients with IBD can have FFM depletion, even in remission [5]. FFM depletion in this population has been associated with osteopenia, primary non-response to infliximab, major postoperative complications, and predicting a small bowel resection [4,7−9]. Therefore, overestimating FFM using unadjusted BIS may mask FFM depletion in this population. Additionally, by overestimating FFM, FM is underestimated. As the IBD population is becoming more overweight and obese, accurate assessments of adiposity will become increasingly important to minimize morbidity [28].

When examining reasons for this bias between methods, an understanding of the underlying theory of bioimpedance analysis is informative. FFM is determined from TBW measurement and relies on an assumption of a constant hydration of the FFM (73.2%), which assumes the individual is normally hydrated [29]. Errors in FFM estimation from bioimpedance are due to errors in TBW estimation and/or an incorrect assumption of the hydration status. Lower TBW values (e.g., dehydration) yield underestimates of FFM; whereas overhydration results in overestimation.

It is theoretically possible that observed overestimations of FFM in patients with IBD arose from BIS overestimating TBW; however, the evidence supporting this is conflicting. In a cohort of patients with gastrointestinal disease, BIS underestimated TBW by 3 L compared with dilution methods [30]. Conversely, overestimation of TBW by BIS compared with dilution techniques was seen in a group of adequately hydrated IBD patients with ileostomies [31].

When considering the other sources of error, the hydration fraction itself has been shown to be stable across healthy populations; however, this has not been studied in all disease states including IBD populations [29]. BIS and MFBIA have theoretical advantages over single-frequency BIA in that the determination of ECW as a percentage of TBW allows the identification of patients with excess ECW (e.g., edema), and thus patients in whom FFM might not truly reflect lean muscle stores. In the present study, the mean ECW% of TBW% was 42%, demonstrating that the cohort was normally hydrated.

BIS has been reported to be less accurate in overweight and obese populations, which is consistent with our findings for unadjusted BIS data [21,32]. Moissl proposed a method to adjust for this by incorporating BMI into prediction algorithms with the aim of improving the body proportion value ($K_b$) [20]. $K_b$ is a factor correcting for a whole-body measurement between wrist and ankle, relating the relative proportions of the leg, arm, trunk, and height used in the Hanai modeling in BIS to determine ECW and therefore FFM [32]. Individual body proportion can be calculated; however, a series of anthropometric measurements of the leg, arm, trunk, and height need to be completed, which is not practical. The default $K_b$ used in BIS SFB7 software is based on population data from US Defence Force personnel in the 1980s [33]. Moissl demonstrated that by adjusting $K_b$ for BMI, the accuracy of FFM prediction could be improved [20]. In the present study, the Moissl approach reduced the mean bias of $FFM_{BIS}$ from 3.4 to 0.1 kg and yielded narrower 95% LOA, making it a good adjustment for this cohort in which 35% (n = 14) was overweight or obese. Equivalence testing between $FFM_{MoissI}$ and $FFM_{DXA}$ at 4.5% suggested the two methods are equivalent and could be used interchangeably while keeping within the 1.5 kg MCID. To operationalize this approach in clinical

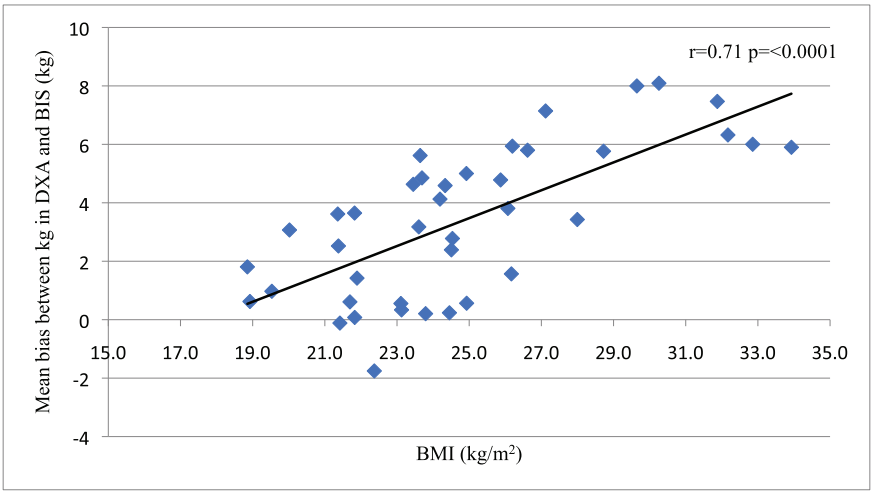


**Fig. 1.** Linear correlation between BMI and mean bias in kg between BIS DXA.

practice, the algorithm would need to be incorporated into the device as an analytical option.

In this sample, the MFBIA estimated FFM more accurately than unadjusted BIS and with lower 95% LOA (−2.6 to 5.2 kg; similar to those of $FFM_{BISMoissl}$) and mean bias of 1.3 ± 2 kg. Mingrone et al. [34] compared FFM by single-frequency BIA to isotopic dilution in CD, showing a mean bias of 0.25 kg; however, that study used a different BIA device to the one used in the present study. To our knowledge, $FFM_{MFBIA}$ has not been compared with $FFM_{DXA}$ in an IBD population. As such, our results represent the first data available demonstrating the performance of MFBIA as a body composition technique in IBD.

Another potential source of error in the bias of the BIA tools was that DXA, although an accepted reference method for body composition, is not without its own limitations, including an approximate 4% LOA against the four-compartment body composition model. This may be a source of part or all of the observed bias in the estimation of FFM [35].

To our knowledge, there is no consensus on the MCID of FFM between body composition tools and reference methods for acceptable levels of mean bias, LOA, or equivalence proportions. This is despite bioimpedance tools being widely used for measuring body composition. Ward [33] has encouraged clinicians to look beyond the "purported poor absolute accuracy" of BIA techniques given the range of errors associated with BIA methods are not dissimilar to those observed for reference standard methods. There is a need to focus on developing a standard definition of MCID between methods, as there is no current consensus on acceptable errors between techniques. Marra et al. [23] proposed that a ±5% FFM could be used to determine accuracy of bioimpedance compared with a reference standard on an individual level. This criterion is not a universally accepted threshold for performance of a bedside body composition at an individual level and it has not yet been assessed for its performance in accurately classifying FFM depletion [10].

Strengths of this study included the evaluation of two bioimpedance devices against DXA as the reference method using standard operating procedures and the evaluation of additional algorithms to improve the accuracy of BIS measurements [20]. The study cohort represented a stable tertiary outpatient IBD population, making the results transferable to other similar populations [5]. One limitation was that results from this study may not be directly generalizable to other bioimpedance devices, each with their own in-built prediction algorithms for body composition, which may be population specific. As a cross-sectional study it was not feasible to evaluate the accuracy of BIS and MFBIA in quantifying changes in FFM occurring over time.

The advantages of bioimpedance methods are clear: Measurements can be done quickly and non-invasively at the bedside or in clinic and without prebooking, do not expose patients to radiation, and can be repeatedly frequently as needed and results are immediately available. Additionally, although the purchase cost of MFBIA and BIS devices is high but less than DXA, the recurrent consumable costs are low (BIS) or absent (seca stand-on style device), making it affordable.

Clinicians could confidently classify FFMI in relation to malnutrition cutoffs, with limited risk for misclassifying patients by using BIS with the Moissl algorithm. MFBIA technologies are likely to perform sufficiently well in the clinical setting provided standardized testing procedures are followed. Users need to be aware of the limitations of the tool, including understanding the contraindications and the width of LOA, which effect accuracy of FFM prediction at the individual level.

*Directions for research*

Studies should validate bioimpedance methods for measuring FFM change in comparison to reference methods. This would allow researchers to develop clinical trials using bioimpedance to investigate the effects of interventions to improve FFM in this population and assess associations between FFM changes and functional and clinical outcomes.

## Conclusion

Given the need for bedside body composition analyses in IBD outpatients, we suggest that MFBIA, or BIS using the Moissl algorithm, are accessible, non-invasive body composition techniques that can be readily incorporated into clinical practice.

## Acknowledgments

The authors acknowledge the study participants and the radiographers who undertook the dual-energy x-ray absorptiometry scans.

## Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.nut.2020.110833.

## References


[1] Mowat C, Cole A, Windsor A, Ahmad T, Arnott I, Driscoll R, et al. Guidelines for the management of inflammatory bowel disease in adults. Gut 2011;60:571–607.
[2] Gong J, Zuo L, Guo Z, Zhang L, Li Y, Gu L, et al. Impact of disease activity on resting energy expenditure and body composition in adult Crohn's Disease: a prospective longitudinal assessment. JPEN J Parenter Enteral Nutr 2015;39:713–8.
[3] Forbes A, Escher J, Hebuterne X, Klek S, Krznaric Z, Schneider S, et al. ESPEN guideline: clinical nutrition in inflammatory bowel disease. Clin Nutr 2017;36:321–47.
[4] Bryant RV, Ooi S, Schultz CG, Goess C, Grafton R, Hughes J, et al. Low muscle mass and sarcopenia: common and predictive of osteopenia in inflammatory bowel disease. Aliment Pharmacol Ther 2015;41:895–906.
[5] Ryan E, McNicholas D, Creavin B, Kelly ME, Walsh T, Beddy D, et al. Sarcopenia and inflammatory bowel disease: a systematic review. Inflamm Bowel Dis 2019;25:67–73.
[6] Schneider SM, Al-Jaouni R, Filippi J, Wiroth JB, Zeanandin G, Arab K, et al. Sarcopenia is prevalent in patients with Crohn's disease in clinical remission. Inflamm Bowel Dis 2008;14:1562–8.
[7] Zhang T, Cao L, Cao T, Yang J, Gong J, Zhu W, et al. Prevalence of sarcopenia and its impact on postoperative outcome in patients with Crohn's disease undergoing bowel resection. JPEN J Parenter Enteral Nutr 2017;41:592–600.
[8] Bamba S, Sasaki M, Takaoka A, Kurihara M, Iwakawa H, Bamba S, et al. Sarcopenia is a predictive factor for intestinal resection in admitted patients with Crohn's disease. PLoS One 2017;12:e0180036.
[9] Ding NS, Malietzis G, Lung PFC, Penez L, Yip WM, Gabe S, et al. The body composition profile is associated with response to anti-TNF therapy in Crohn's disease and may offer an alternative dosing paradigm. Aliment Pharmacol Ther 2017;46:883–91.
[10] Cederholm T, Bosaeus I, Barazzoni R, Bauer J, Van Gossum A, Klek S, et al. Diagnostic criteria for malnutrition—an ESPEN Consensus Statement. Clin Nutr 2015;34:335–40.
[11] Cederholm T, Jensen GL, Correia MITD, Gonzalez MC, Fukushima R, Higashiguchi T, et al. GLIM criteria for the diagnosis of malnutrition—a consensus report from the global clinical nutrition community. Clin Nutr 2019;38:1–9.
[12] Yadav DP, Kedia S, Madhusudhan KS, Bopanna S, Goyal S, Jain S, et al. Body composition in Crohn's disease and ulcerative colitis: correlation with disease severity and duration. Can J Gastroenterol Hepatol 2017:1215035.
[13] Wiech P, Binkowska-Bury M, Korczowski B. Body composition as an indicator of the nutritional status in children with newly diagnosed ulcerative colitis and Crohn's disease—a prospective study. Prz Gastroenterol 2017;12:55–9.
[14] Machado JF, Oya V, Coy CS, Morcillo AM, Severino SD, Wu C, et al. Whey and soy protein supplements changes body composition in patients with Crohn's disease undergoing azathioprine and anti-TNF-alpha therapy. Nutr Hosp 2015;31:1603–10.
[15] Schroeder KW, Tremaine WJ, Ilstrup DM. Coated oral 5-aminosalicylic acid therapy for mildly to moderately active ulcerative colitis. A randomized study. N Engl J Med 1987;317:1625–9.
[16] Harvey R, Bradshaw J. A simple index of Crohn's-disease activity. Lancet 1980;315:514.
[17] Satsangi J, Silverberg MS, Vermeire S, Colombel JF. The Montreal classification of inflammatory bowel disease: controversies, consensus, and implications. Gut 2006;55:749–53.
[18] World Health Organization. Obesity: preventing and managing the global epidemic. Report of a WHO consultation. World Health Organ Tech Rep Ser 2000;894:1–253. i-xii.
[19] Ward LC, Isenring E, Dyer JM, Kagawa M, Essex T. Resistivity coefficients for body composition analysis using bioimpedance spectroscopy: effects of body dominance and mixture theory algorithm. Physiol Meas 2015;36:1529–49.
[20] Moissl UM, Wabel P, Chamney PW, Bosaeus I, Levin NW, Bosy-Westphal A, et al. Body fluid volume determination via body composition spectroscopy in health and disease. Physiol Meas 2006;27:921–33.
[21] Ward L, Dyer J, Byrne N, Sharpe KK, Hills AP. Validation of a three-frequency bioimpedance spectroscopic method for body composition analysis. Nutrition 2007;23:657–63.
[22] Giavarina D. Understanding Bland Altman analysis. Biochem Med (Zagreb) 2015;25:141–51.
[23] Marra M, Sammarco R, De Filippo E, Caldara A, Speranza E, Scalfi L, et al. Prediction of body composition in anorexia nervosa: results from a retrospective study. Clin Nutr 2018;37:1670–4.
[24] Ellegard L, Bertz F, Winkvist A, Bosaeus I, Brekke HK. Body composition in overweight and obese women postpartum: bioimpedance methods validated by dual energy x-ray absorptiometry and doubly labeled water. Eur J Clin Nutr 2016;70:1181–8.
[25] Birzniece V, Khaw CH, Nelson AE, Meinhardt U, Ho KK. A critical evaluation of bioimpedance spectroscopy analysis in estimating body composition during GH treatment: comparison with bromide dilution and dual X-ray absorptiometry. Eur J Endocrinol 2015;172:21–8.
[26] Ellegard LH, Ahlen M, Korner U, Lundholm KG, Plank LD, Bosaeus IG. Bioelectric impedance spectroscopy underestimates fat-free mass compared to dual energy x-ray absorptiometry in incurable cancer patients. Eur J Clin Nutr 2009;63:794–801.
[27] Hronek M, Kovarik M, Aimova P, Koblizek V, Pavlikova L, Salajka F, et al. Skinfold anthropometry—the accurate method for fat free mass measurement in COPD. COPD 2013;10:597–603.
[28] Bryant RV, Schultz CG, Ooi S, Goess C, Costello SP, Vincent AD, et al. Obesity in inflammatory bowel disease: gains in adiposity despite high prevalence of myopenia and osteopenia. Nutrients 2018;10:1192.
[29] Wang Z, Deurenberg P, Wang W, Pietrobelli A, Baumgartner RN, Heymsfield SB. Hydration of fat-free body mass: review and critique of a classic body-composition constant. Am J Clin Nutr 1999;69:833–41.
[30] Cox-Reijven PL, van Kreel B, Soeters PB. Bioelectrical impedance measurements in patients with gastrointestinal disease: validation of the spectrum approach and a comparison of different methods for screening for nutritional depletion. Am J Clin Nutr 2003;78:1111–9.
[31] Carlsson E, Bosaeus I, Nordgren S. Body composition in patients with an ileostomy and inflammatory bowel disease: validation of bio-electric impedance spectroscopy (BIS). Eur J Clin Nutr 2002;56:680–6.
[32] Cox-Reijven PL, Soeters PB. Validation of bio-impedance spectroscopy: effects of degree of obesity and ways of calculating volumes from measured resistance values. Int J Obes Relat Metab Disord 2000;24:271–80.
[33] Ward LC. Bioelectrical impedance analysis for body composition assessment: reflections on accuracy, clinical utility, and standardisation. Eur J Clin Nutr 2019;73:194–9.
[34] Mingrone G, Benedetti G, Capristo E, De Gaetano A, Greco AV, Tataranni PA, et al. Twenty-four-hour energy balance in Crohn disease patients: metabolic implications of steroid treatment. Am J Clin Nutr 1998;67:118.
[35] Van Der Ploeg GE, Withers RT, Laforgia J. Percent body fat via DEXA: comparison with a four-compartment model. J Appl Physiol 2003;94:499–506. (1985).